

ACADEMIC REGULATIONS & SYLLABUS

**Faculty of Pharmacy
Master of Pharmacy Programme
(Pharmaceutical Chemistry)**

(AS PER PCI SYLLABUS)

A.Y. 2023-2024

Ramanbhai Patel College of Pharmacy

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (M. Pharm.) PROGRAMME

Vision of RPCP

*To Become a Premier Pharma Institute by Creating World Class Pharmacists
and Researchers.*

Mission of RPCP

*To Strive for the Excellence in Pharmaceutical Sciences through Quality
Education and Research.*

PROGRAM OUTCOMES

- 1. Pharmacy Knowledge:** Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.
- 2. Planning Abilities:** Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.
- 3. Problem analysis:** Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.
- 4. Modern tool usage:** Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.
- 5. Leadership skills:** Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfilment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.
- 6. Professional Identity:** Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).
- 7. Pharmaceutical Ethics:** Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.
- 8. Communication:** Communicate effectively with the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
- 9. The Pharmacist and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.
- 10. Environment and sustainability:** Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 11. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

FACULTY OF PHARMACY
ACADEMIC REGULATIONS
MASTER OF PHARMACY (M. Pharm.) PROGRAMME
Choice Based Credit System (CBCS)

1. Short Title and Commencement

These regulations shall be called as “The Revised Academic Regulations for the postgraduate programmes under the Faculty of Pharmacy”. They shall come into effect from the Academic Year 2018-19. The regulations framed are subject to modifications from time to time by the respective regulatory bodies.

2. Minimum Qualification for Admission

2.1 Candidate shall have passed B. Pharm Degree examination of an Indian university established by law in India from an institution approved by Pharmacy Council of India and has scored not less than 55 % of the maximum marks (aggregate of 4 years of B.Pharm.)

2.2 Every student, selected for admission to post graduate pharmacy program in any PCI approved institution should have obtained registration with the State Pharmacy Council or should obtain the same within one month from the date of his/her admission, failing which the admission of the candidate shall be cancelled.

Note: It is mandatory to submit a migration certificate obtained from the respective university where the candidate had passed his/her qualifying degree (B.Pharm.).

3. Duration of the Programme

The course of study for M.Pharm shall extend over a period of four semesters (two academic years). The curricula and syllabi for the program shall be prescribed from time to time by Pharmacy Council of India, New Delhi.

4. Medium of Instruction and Examinations

Medium of instruction and examination shall be in English.

5. Working Days in a Semester

Each semester shall consist of not less than 100 working days. The odd semesters shall be conducted from the month of June/July to November/December and the even semesters shall be conducted from December/January to May/June in every calendar year.

6. Attendance and Progress

A candidate is required to put in at least 80% attendance in individual courses considering theory and practical separately. The candidate shall complete the prescribed course satisfactorily to be eligible to appear for the respective examinations.

7. Programme Credit Structure

As per the philosophy of Credit Based Semester System, certain quantum of academic work viz. theory classes, tutorial hours, practical classes, etc. are measured in terms of credits. On satisfactory completion of the courses, a candidate earns credits. The amount of credit associated with a course is dependent upon the number of hours of instruction per week in that course. Similarly, the credit associated with any of the other academic, co/extra-curricular activities is dependent upon the quantum of work expected to be put in for each of these activities per week.

7.1. Credit assignment

7.1.1. Theory and Laboratory courses

Courses are broadly classified as Theory and Practical. Theory courses consist of lecture (L) and Practical (P) courses consist of hours spent in the laboratory. Credits (C) for a course is dependent on the number of hours of instruction per week in that course, and is obtained by using a multiplier of one (1) for lecture and a multiplier of half (1/2) for practical (laboratory) hours. Thus, for example, a theory course having four lectures per week throughout the semester carries a credit of 4. Similarly, a practical having four laboratory hours per week throughout semester carries a credit of 2.

The contact hours of seminars, assignments and research work shall be treated as that of practical courses for the purpose of calculating credits. i.e., the contact hours shall be multiplied by 1/2. Similarly, the contact hours of journal club, research work presentations and discussions with the supervisor shall be considered as theory course and multiplied by 1.

7.2. Minimum credit requirements

The minimum credit points required for the award of M. Pharm. degree is 93. However based on the credit points earned by the students under the head of co-curricular activities, a student shall earn a maximum of 98 credit points. These credits are divided into Theory courses, Practical, Seminars, Assignments, Research work, Discussions with the supervisor, Journal club and Co-Curricular activities over the duration of four semesters. The credits are distributed semester-wise. Courses generally progress in sequence, building competencies and their positioning

indicates certain academic maturity on the part of the learners. Learners are expected to follow the semester-wise schedule of courses given in the syllabus.

8. Academic work

A regular record of attendance both in Theory, Practical, Seminar, Assignment, Journal club, Discussion with the supervisor, Research work presentation and Dissertation shall be maintained by the department / teaching staff of respective courses.

9. Examinations/Assessments

The scheme for internal assessment and end semester examinations is given in Annexure II (Table – 1 to 5).

9.1. End Semester Examinations

The End Semester Examinations for each theory and practical course through semesters I to IV shall be conducted by the University for which Examinations shall be conducted by the subject experts at college level and the marks/grades shall be submitted to the university.

9.2. Internal Assessment: Continuous Mode

The marks allocated for Continuous mode of Internal Assessment shall be awarded as per the scheme given below.

Table- 1: Scheme for awarding internal assessment: Continuous mode

Theory	
Criteria	Maximum Marks
Attendance (Refer Table – 2)	8
Student – Teacher interaction	2
Total	10
Practical	
Attendance (Refer Table – 2)	10
Based on Practical Records, Regular viva voce, etc.	10
Total	20

Table- 2: Guidelines for the allotment of marks for attendance

Percentage of Attendance	Theory	Practical
95 – 100	8	10
90 – 94	6	7.5
85 – 89	4	5
80 – 84	2	2.5
Less than 80	0	0

9.2.1. Sessional Exams

Two Sessional exams shall be conducted for each theory / practical course as per the schedule fixed by the college. The scheme of question paper for theory and practical Sessional examinations will be as prescribed by the regulatory body. The average marks of two Sessional exams shall be computed for internal assessment as per the requirements given in Annexure II. Sessional exam shall be conducted for 30 marks for theory and shall be computed for 15 marks. Similarly, Sessional exam for practical shall be conducted for 40 marks and shall be computed for 10 marks.

10. Promotion and Award of Grades

A student shall be declared PASS and eligible for getting grade in a course of M.Pharm. Programme, if he/she secures at least 50% marks in that particular course including internal assessment.

11. Carry Forward of Marks

In case a student fails to secure the minimum 50% in any Theory or Practical course as specified in 9 above, then he/she shall reappear for the end semester examination of that course. However, his/her marks of the Internal Assessment shall be carried over and he/she shall be entitled for grade obtained by him/her on passing.

12. Improvement of Internal Assessment

A student shall have the opportunity to improve his/her performance only once in the Sessional exam component of the internal assessment. The re-conduct of the Sessional exam shall be completed before the commencement of next end semester theory examinations.

13. Re-examination of End Semester Examinations

Re-examination of end semester examination shall be conducted as per the schedule given in table 3. The exact dates of examinations shall be notified from time to time.

Table-3: Tentative schedule of end semester examinations

Semester	For Regular Candidates	For Failed Candidates	
		Remedial Examination-1	Remedial Examination-2*
I and III	November / December	After 15 days of the declaration of the result of the end semester Examination	last week of June/

II and IV	May / June	After 15 days of the declaration of the result of the end semester Examination	end of the Academic Year, which is earlier
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* If student who are remain left from end semester examination and remedial examination-1.

14. Academic Progression

No student shall be admitted to any examination unless he/she fulfils the norms given in item no. 6 under the heading of attendance and progress. Academic progression rules are applicable as follows:

- “Student will not be allowed to move to next year if she/he has not cleared all the courses of the previous year”
- A student shall be eligible to carry forward all the courses of Ist semesters till the IInd semester examinations. However, he/she shall not be eligible to attend the courses of the IIIrd semester until all the courses of the Ist and IInd semesters are successfully completed.
- A student shall be eligible to get his/her CGPA upon successful completion of the courses of I to IV semesters within the stipulated time period as per the norms.
- Note: Grade “NA” should be considered as failed and treated as one head for deciding ATKT. Such rules are also applicable for those students who fail to register for examination(s) of any course in any semester.

15. Grading of Performances (Letter Grades and Grade Points Allocations)

Based on the performances, each student shall be awarded a final letter grade at the end of the semester for each course. The letter grades and their corresponding grade points are given in Table – 4.

Table-4: Letter grades and grade points equivalent to Percentage of marks and performances

Percentage of Marks Obtained	Letter Grade	Grade Point	Performance
90.00 – 100	AA	10	Outstanding
80.00 – 89.99	AB	9	Excellent
70.00 – 79.99	BB	8	Good
60.00 – 69.99	BC	7	Fair
50.00 – 59.99	CC	6	Average

Less than 50	FF	0	Fail
Absent	NA	0	Fail

A learner who remains absent for any end semester examination shall be assigned a letter grade of “NA” and a corresponding grade point of zero. He/she should reappear for the said evaluation/examination in due course.

16. Semester Grade Point Average (SGPA)

The performance of a student in a semester is indicated by a number called ‘Semester Grade Point Average’ (SGPA). The SGPA is the weighted average of the grade points obtained in all the courses by the student during the semester. For example, if a student takes five courses(Theory/Practical) in a semester with credits C1, C2, C3, C4 and C5 and the student’s grade points in these courses are G1, G2, G3, G4 and G5, respectively, and then students’ SGPA is equal to:

$$SGPA = \frac{C1G1+C2G2+C3G3+C4G4}{C1+C2+C3+C4}$$

The SGPA is calculated to two decimal points. It should be noted that, the SGPA for any semester shall take into consideration the FF and Ab. grade awarded in that semester. For example if a learner has a FF or Ab. grade in course 4, the SGPA shall then be computed as:

$$SGPA = \frac{C1G1+C2G2+C3G3+C4Zero}{C1+C2+C3+C4}$$

17. Cumulative Grade Point Average (CGPA)

The CGPA is calculated with the SGPA of all the VIII semesters to two decimal points and is indicated in final grade report card/final transcript showing the grades of all VIII semesters and their courses. The CGPA shall reflect the failed status in case of FF grade(s), till the course(s) is/are passed. When the course(s) is/are passed by obtaining a pass grade on subsequent examination(s) the CGPA shall only reflect the new grade and not the fail grades earned earlier. The CGPA is calculated as:

$$CGPA = \frac{C1S1+C2S2+C3S3+C4S4}{C1+C2+C3+C4}$$

where C1, C2, C3,... is the total number of credits for semester I,II,III,... and S1,S2, S3,...is the SGPA of semester I,II,III,... .

No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

18. Declaration of Class (Table-5)

The class shall be awarded on the basis of CGPA as follows:

First Class with Distinction	= CGPA of. 7.50 to 10.0
First Class	= CGPA of 6.0 to 7.49
Second Class	= CGPA of 5.0 to 5.99
Pass Class	< CGPA of 5.00

19. Project Work

All the students shall undertake a project under the supervision of a teacher in Semester III to IV and submit a report. 4 copies of the project report shall be submitted (typed & bound copy not less than 75 pages).

The internal and external examiner appointed by the University shall evaluate the project at the time of the Practical examinations of other semester(s). The projects shall be evaluated as per the criteria given below.

Evaluation of Dissertation Book				
Criteria	Semester-III		Semester-IV	
			Internal Evaluation (Marks)	External Evaluation (Marks)
Objective(s) of the work done	--		05	05
Methodology Adopted	--		25	25
Results and Discussions	--		15	15
Conclusions and Outcome	--		05	05
Total	--		50	50
Final Total			100	
Evaluation of Presentation				
Criteria	Semester-III		Semester-IV	
	Internal Evaluation (Marks)	External Evaluation (Marks)	Internal Evaluation (Marks)	External Evaluation (Marks)
Presentation of work	75	100	75	75
Communication skills	25	50	25	25
Question and answer skills	50	50	50	50
Total	150	200	150	150
Final Total	350		300	

20. Award of Ranks

Ranks and Medals shall be awarded on the basis of final CGPA. However, candidates who fail in one or more courses during the M.Pharm program shall not be eligible for award of ranks. Moreover, the candidates should have completed the M. Pharm program in minimum prescribed number of years, (two years) for the award of Ranks.

22. Award of degree

Candidates who fulfil the requirements mentioned above shall be eligible for award of degree during the ensuing convocation.

23. Duration for completion of the program of study

The duration for the completion of the program shall be fixed as double the actual duration of the program and the students have to pass within the said period, otherwise they have to get fresh Registration.

24. Extra Credit:

An extra credit is to be offered to a student for achievements in co-curricular and extra-curricular activities. This credit shall not be counted while considering the minimum credits for completing the program. The activities and appropriate weight (points) to be allocated to award an extra credit are broadly classified as per the table below:

Sr. no.	Name of the Activity	Maximum Credit Points Eligible / Activity
1	Participation in National Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student)	01
2	Participation in international Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student)	02
3	Academic Award/Research Award from State Level/National Agencies	01
4	Academic Award/Research Award from International Agencies	02
5	Research / Review Publication in National Journals (Indexed in Scopus / Web of Science)	01
6	Research / Review Publication in International Journals (Indexed in Scopus / Web of Science)	02

Note: International Conference: Held Outside India

International Journal: The Editorial Board Outside India

***The credit points assigned for extracurricular and or co-curricular activities shall be given by the Principals of the colleges and the same shall be submitted to the University. The criteria to acquire this credit point shall be defined by the colleges from time to time.**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME

Schemes for internal assessments and end semester examinations

SEMESTER-I
SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC009	Modern Pharmaceutical Analytical Techniques	4	4	4
PHPCM001	Advanced Organic Chemistry-I	4	4	4
PHPCM002	Advanced Medicinal Chemistry	4	4	4
PHPCM003	Chemistry of Natural Products	4	4	4
PHPCM004	Pharmaceutical Chemistry Practical – I	12	6	12
PHPCM005	Seminar/Assignment-I	2	1	2
---	DHSS Elective-I*	2	2	2
Total		32	25	32

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHCCC009	Modern Pharmaceutical Analytical Techniques	10	15	1Hr	25	75	3Hrs	100
PHPCM001	Advanced Organic Chemistry-I	10	15	1Hr	25	75	3Hrs	100
PHPCM002	Advanced Medicinal Chemistry	10	15	1Hr	25	75	3Hrs	100
PHPCM003	Chemistry of Natural Products	10	15	1Hr	25	75	3Hrs	100
PHPCM004	Pharmaceutical Chemistry Practical - I	20	30	6Hrs	50	100	6Hrs	150
PHPCM005	Seminar/Assignment-I	-	-	-	100	-	-	100
---	DHSS elective-I*	-	-	-	30	70	-	100
Total								750

***DHSS elective courses: SCHEME OF TEACHING**

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
I	HS105.02 B	Academic Speaking and Presentation Skills	02	02	02

***DHSS elective courses: SCHEME OF EVALUATION**

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS105.02 B	Academic Speaking and Presentation Skills	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-II
SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHPCM006	Advanced Spectral Analysis	4	4	4
PHPCM007	Advanced Organic Chemistry-II	4	4	4
PHPCM008	Computer Aided Drug Design	4	4	4
PHPCM009	Pharmaceutical Process Chemistry	4	4	4
PHPCM010	Pharmaceutical Chemistry Practical-II	12	6	12
PHPCM011	Seminar/Assignment-II	2	1	2
---	DHSS elective-II*	2	2	2
---	University Elective-II**	2	2	2
Total		34	27	34

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPCM006	Advanced Spectral Analysis	10	15	1Hr	25	75	3Hrs	100
PHPCM007	Advanced Organic Chemistry-II	10	15	1Hr	25	75	3Hrs	100
PHPCM008	Computer Aided Drug Design	10	15	1Hr	25	75	3Hrs	100
PHPCM009	Pharmaceutical Process Chemistry	10	15	1Hr	25	75	3Hrs	100
PHPCM010	Pharmaceutical Chemistry Practical-II	20	30	6Hrs	50	100	6Hrs	150
PHPCM011	Seminar/Assignmen t-II	-	-	-	100	-	-	100
---	DHSS elective-II*	-	-	-	30	70	-	100
---	University Elective-II**	-	-	-	30	70	-	100
Total								850

***DHSS elective courses: SCHEME OF TEACHING**

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
II	HS106.02 B	Academic Writing	02	02	02

***DHSS elective courses: SCHEME OF EVALUATION**

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS106.02 B	Academic Writing	-	-	30	70	100

****University elective courses: SCHEME OF TEACHING - Semester-II**

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
CE771.01	Project Management	2	2	2
ME781.01	Occupational Health and Safety	2	2	2
CA730	Internet and Web Designing	2	2	2
CA842.01	Mobile Application Development	2	2	2
PT796.01	Fitness and Nutrition	2	2	2
MB651	Software Based Statistical Analysis	2	2	2
NR755	First Aid and Life Support	2	2	2
OC733.01	Introduction to Polymer Science	2	2	2
PH892.01	Intellectual Property Rights	2	2	2
PH893	Clinical Data & Interpretation	2	2	2

****University elective courses: SCHEME OF EVALUATION- Semester-II**

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
CE771.01	Project Management	-	-	30	70	100
ME781.01	Occupational Health and Safety	-	-	30	70	100
CA730	Internet and Web Designing	-	-	30	70	100
CA842.01	Mobile Application Development	-	-	30	70	100
PT796.01	Fitness and Nutrition	-	-	30	70	100
MB651	Software Based Statistical Analysis	-	-	30	70	100
NR755	First Aid and Life Support	-	-	30	70	100
OC733.01	Introduction to Polymer Science	-	-	30	70	100
PH892.01	Intellectual Property Rights	-	-	30	70	100
PH893	Clinical Data & Interpretation	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-III

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC010	Research Methodology and Biostatistics*	4	4	4
PHCCC011	Journal Club-I	01	01	01
PHPCM012	Discussion / Presentation (Proposal Presentation)	02	02	02
PHPCM013	Research Work-I	28	14	28
	Total	35	21	35

***Non University Examination**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHCCC010	Research Methodology and Biostatistics	10	15	1	25	75	3	100
PHPCM012	Discussion / Presentation (Proposal Presentation)	-	-	-	50	-	-	50
PHCCC011	Journal Club-I	-	-	-	25	-	-	25
PHPCM013	Research Work-I				-	350	1hr	350
Total								525

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER -IV

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC012	Journal Club-II	1	1	1
PHPCM015	Research Work-II	31	16	31
PHPCM014	Discussion/Presentation	03	03	03
Total		35	20	35

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPCM014	Discussion/ Presentation	-	-	-	75	-	-	75
PHCCC012	Journal Club-II	-	-	-	25	-	-	25
PHPCM015	Research Work-II				-	400	1hr	400
Total								500

Semester	Credit Point
I	25
II	27
III	21
IV	20
Co-curricular Activity (Attending Conference, Scientific Presentations, Publication, Industry Training and other scholarly Activities)	Maximum 5
Total Credit Point	Minimum- 93 Maximum-98*

***Credit Point for co-curricular activity**

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester I

Charotar University of Science and Technology

FACULTY OF PHARMACY
BACHELOR OF PHARMACY PROGRAMME
PHCCC009: MODERN PHARMACEUTICAL ANALYTICAL
TECHNIQUES (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	UV-Visible spectroscopy	10
	IR spectroscopy	
	Spectrofluorimetry	
	Flame emission spectroscopy and Atomic absorption spectroscopy	
2	NMR Spectroscopy	10
3	Mass Spectroscopy	10
4	Chromatography	10
5	Electrophoresis	10
	X-Ray Crystallography	
6	Potentiometry	10
	Thermal Techniques	

Detailed Syllabus:

1.	UV-Visible spectroscopy	10 Hours	16.67%
	UV-Visible spectroscopy: Introduction, Theory, Laws, Instrumentation associated with UV-Visible spectroscopy, Choice of solvents and solvent effect and Applications of UV Visible spectroscopy.		
	IR spectroscopy		
	Theory, Modes of Molecular vibrations, Sample handling, Instrumentation of Dispersive and Fourier - Transform IR Spectrometer, Factors affecting vibrational frequencies and		

	Applications of IR spectroscopy		
	Spectrofluorimetry		
	Theory of Fluorescence, Factors affecting fluorescence, Quenchers, Instrumentation and Applications of fluorescence spectrophotometer.		
	Flame emission spectroscopy and Atomic absorption spectroscopy		
	Principle, Instrumentation, Interferences and Applications.		
2.	NMR Spectroscopy	10 Hours	16.67%
	Quantum numbers and their role in NMR, Principle, Instrumentation, Solvent requirement in NMR, Relaxation process, NMR signals in various compounds, Chemical shift, Factors influencing chemical shift, Spin-Spin coupling, Coupling constant, Nuclear magnetic double resonance, Brief outline of principles of FT-NMR and ¹³C NMR. Applications of NMR spectroscopy.		
3.	Mass Spectroscopy	10 Hours	16.67%
	Mass Spectrometry: Principle, Theory, Instrumentation of Mass Spectrometry, Different types of ionization like electron impact, chemical, field, FAB and MALDI, APCI, ESI, APPI Analysers of Quadrupole and Time of Flight, Mass fragmentation and its rules, Meta stable ions, Isotopic peaks and Applications of Mass spectrometry.		
4.	Chromatography	10 Hours	16.67%
	Principle, apparatus, instrumentation, chromatographic parameters, factors affecting resolution and applications of the following: a) Thin Layer chromatography b) High Performance Thin Layer Chromatography c) Ion exchange chromatography d) Column chromatography e) Gas chromatography f) High Performance Liquid chromatography g) Ultra High Performance Liquid chromatography h) Affinity		

	chromatography i) Gel Chromatography		
5.	Electrophoresis	10 Hours	16.67%
	Principle, Instrumentation, Working conditions, factors affecting separation and applications of the following: a) Paper electrophoresis b) Gel electrophoresis c) Capillary electrophoresis d) Zone electrophoresis e) Moving boundary electrophoresis f) Isoelectric focusing		
	X-ray Crystallography		
	Production of X-rays, Different X-ray diffraction methods, Bragg's law, Rotating crystal technique, X-ray powder technique, Types of crystals and applications of X-ray diffraction.		
6.	Potentiometry	10 Hours	16.67%
	Principle, working, Ion selective Electrodes and Application of potentiometry		
	Thermal Techniques		
	- Differential Scanning Calorimetry (DSC): Principle, thermal transitions and Instrumentation (Heat flux and power-compensation and designs), Modulated DSC, Hyper DSC, experimental parameters (sample preparation, experimental conditions, calibration, heating and cooling rates, resolution, source of errors) and their influence, advantage and disadvantages, pharmaceutical applications. - Differential Thermal Analysis (DTA): Principle, instrumentation and advantage and disadvantages, pharmaceutical applications, derivative differential thermal analysis (DDTA). TGA: Principle, instrumentation, factors affecting results, advantage and disadvantages, pharmaceutical applications.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe theory and principle of various spectroscopic and chromatographic separation techniques.
CO2	describe instrumentation and application of various spectroscopic and chromatographic separation techniques with justification.
CO3	summarize approaches to be adopted for quantitative & qualitative analysis of drugs in single and combine dosage forms.
CO4	Describe use of thermal methods and potentiometry in analysis of drugs/formulations
CO5	interpret the spectra and propose the structure of organic compounds.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	3	-	3	3
CO3	3	3	3	3	-	3	-	3	-	3	3
CO4	3	3	3	3	-	-	-	3	-	-	3
CO5	3	-	3	3	-	-	-	3	-	-	3

❖ References:

1. Spectrometric Identification of Organic compounds - Robert M Silverstein, Sixth edition, John Wiley & Sons, 2004.
2. Principles of Instrumental Analysis - Douglas A Skoog, F. James Holler, Timothy A. Nieman, 5th edition, Eastern press, Bangalore, 1998
3. Instrumental methods of analysis – Willards, 7th edition, CBS publishers.
4. Practical Pharmaceutical Chemistry – Beckett and Stenlake, Vol II, 4th edition, CBS Publishers, New Delhi, 1997.
5. Organic Spectroscopy - William Kemp, 3rd edition, ELBS, 1991.
6. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi, 3rd Edition, CBS Publishers, New Delhi, 1997.

7. Pharmaceutical Analysis- Modern methods – Part B - J W Munson, Volume 11, Marcel Dekker Series.
8. The Analysis of Drugs in Biological Fluids, Joseph Chamberlain, CRC Press

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM001: ADVANCED ORGANIC CHEMISTRY-I (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Aspects of Organic Chemistry	12
2	Study of mechanism and synthetic applications of named Reactions	12
3	Synthetic Reagents & Applications and Protection-Deprotection of functional group	12
4	Heterocyclic Chemistry	12
5	Synthon approach and retrosynthesis applications	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Basic Aspects of Organic Chemistry	12	20%
	a. Organic intermediates: Carbocations, carbanions, free radicals, carbenes and nitrenes. Their method of formation, stability and synthetic applications. b. Types of reaction mechanisms and methods of determining them c. Detailed knowledge regarding the reactions, mechanisms and their relative reactivity and orientations. d. Nucleophilic uni- and bimolecular reactions (SN ₁ and SN ₂) Elimination reactions (E1 & E2; Hoffman & Saytzeff's rule) Rearrangement reaction		
2.	Study of mechanism and synthetic applications of following named reactions	12	20%
	Ugi reaction, Brook rearrangement, Ullmann coupling reactions, Dieckmann Reaction, Doebner-Miller Reaction, Sandmeyer		

	Reaction, Mitsunobu reaction, Mannich reaction, Vilsmeier-Haack Reaction, Sharpless asymmetric epoxidation, Baeyer-Villiger oxidation, Shapiro & Suzuki reaction, Ozonolysis and Michael addition reaction		
3.	Synthetic Reagents & Applications	12	20%
	Aluminiumisopropoxide, N-bromosuccinamide, diazomethane, dicyclohexylcarbodiimide, Wilkinson reagent, Witting reagent. Osmium tetroxide, titanium chloride, diazopropane, diethyl azodicarboxylate, Triphenylphosphine, Benzotriazol-1-yloxy) tris (dimethylamino) phosphonium hexafluoro-phosphate (BOP). 3.1 Protecting groups Role of protection in organic synthesis Protection for the hydroxyl group, including 1,2-and 1,3-diols: ethers, esters, carbonates, cyclic acetals & ketals Protection for the Carbonyl Group: Acetals and Ketals Protection for the Carboxyl Group: amides and hydrazides, esters Protection for the Amino Group and Amino acids: carbamates and amides		
4.	Heterocyclic Chemistry	12	20%
	4.1 Organic Name reactions with their respective mechanism and application involved in synthesis of drugs containing five, six membered and fused heterocyclic such as Debus-Radziszewski imidazole synthesis, Knorr Pyrazole Synthesis Pinner Pyrimidine Synthesis, Combes Quinoline Synthesis, Bernthsen Acridine Synthesis, Smiles rearrangement and Traube purine synthesis. 4.2 Synthesis of few representative drugs containing these heterocyclic nucleus such as Ketoconazole, Metronidazole, Miconazole, celecoxib, antipyrin, Metamizole sodium, Terconazole, Alprazolam, Triamterene, Sulfamerazine, Trimethoprim, Hydroxychloroquine, Quinine, Chloroquine,		

	Quinacrine, Amsacrine, Prochlorperazine, Promazine, Chlorpromazine, Theophylline, Mercaptopurine and Thioguanine.		
5.	Synthon approach and retrosynthesis applications	12	20%
	a. Basic principles, terminologies and advantages of retrosynthesis; guidelines for dissection of molecules. b. Functional group interconversion and addition (FGI and FGA) c. C-X disconnections; C-C disconnections – alcohols and carbonyl compounds; 1,2-, 1,3-,1,4-, 1,5-, 1,6-difunctionalized compounds d. Strategies for synthesis of three, four, five and six-membered ring.		

Course Outcome (COs):

CO1	describe the mechanism & applications of various named reactions
CO2	understand the chemistry of heterocyclic compounds
CO3	narrate concept of disconnection to develop synthetic routes for small target molecule.
CO4	summarize various catalysts used in organic reactions

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	3	3

Recommended Study Material:

❖ Text book:

1. “Advanced Organic chemistry, Reaction, Mechanisms and Structure”, J March, John Wiley and Sons, New York.
2. “Mechanism and Structure in Organic Chemistry”, ES Gould, Hold Rinchart and Winston, New York.

3. "Organic Chemistry" Clayden, Greeves, Warren and Wothers., Oxford University Press 2001.
4. "Organic Chemistry" Vol I and II. I.L. Finar. ELBS, Pearson Education Ltd, Dorling Kindersley 9 India) Pvt. Ltd.
5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
6. Reactive Intermediates in Organic Chemistry, Tandon and Gower, Oxford & IBH Publishers.
7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.
8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wiley India
10. Principles of Organic Synthesis, R.C. Norman and J.M. Coxon, Nelson Thornes.
11. Organic Synthesis - Special Techniques. VK Ahluwalia and R Agarwal, Narosa Publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM002: ADVANCED MEDICINAL CHEMISTRY (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Drug Discovery and Drug Target	12
2	Prodrug Design and Analog design	12
3	Advanced Medicinal Chemistry of Selective Class of Drug	12
4	Rational Design of Enzyme Inhibitors	12
5	Peptidomimetics	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Drug Discovery and Drug Target	12	20%
	Drug discovery: Stages of drug discovery, lead discovery; identification, validation and diversity of drug targets. Biological drug targets: Receptors, types, binding and activation, theories of drug receptor interaction, drug receptor interactions, agonists vs antagonists, artificial enzymes.		
2.	Prodrug Design and Analog design	12	20%
	2.1 Prodrug design: Basic concept, Carrier linked prodrugs/ Bioprecursors, Prodrugs of functional group, Prodrugs to improve patient acceptability, Drug solubility, Drug absorption and distribution, site specific drug delivery and sustained drug action. Rationale of prodrug design and practical consideration of prodrug design. 2.2 Combating drug resistance: Causes for drug resistance,		

	strategies to combat drug resistance in antibiotics and anticancer therapy, Genetic principles of drug resistance. C 2.3 Analog Design: Introduction, Classical & Non classical, Bioisosteric replacement strategies, rigid analogs, alteration of chain branching, changes in ring size, ring position isomers, design of stereo isomers and geometric isomers, fragments of a lead molecule, variation in inter atomic distance.		
3.	Advanced Medicinal Chemistry of Selective Class of Drug	12	20%
	Systematic study, SAR, Mechanism of action and synthesis of new generation molecules of following class of drugs: 3.1 Anti-hypertensive drugs, Psychoactive drugs, Anticonvulsant drugs, H1 & H2 receptor antagonist, COX1 & COX2 inhibitors, Adrenergic & Cholinergic agents, Antineoplastic and Antiviral agents. 3.2 Stereochemistry and Drug action: Realization that stereo selectivity is a pre-requisite for evolution. Role of chirality in selective and specific therapeutic agents. Case studies, Enantio-selectivity in drug adsorption, metabolism, distribution and elimination.		
4.	Rational Design of Enzyme Inhibitors	12	20%
	Enzyme kinetics & Principles of Enzyme inhibitors, Enzyme inhibitors in medicine, Enzyme inhibitors in basic research, rational design of non-covalently and covalently binding enzyme inhibitors.		
5.	Peptidomimetics	12	20%
	Therapeutic values of Peptidomimetics, design of peptidomimetics by manipulation of the amino acids, modification of the peptide backbone, incorporating conformational constraints locally or globally. Chemistry of prostaglandins, leukotrienes and thromboxones.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	Describe different stages and techniques of Drug Discovery and Development
CO2	Summarize various strategies to design and develop new drug like molecules for biological targets
CO3	Understand role of medicinal chemistry in drug research
CO4	Narrate designing strategy and therapeutical value of peptidomimetics.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	3	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	3	-	-	-	-	-	-	3

Recommended Study Material:**❖ Text book:**

1. Medicinal Chemistry by Burger, Vol I –VI. 2.
2. Wilson and Gisvold's Text book of Organic Medicinal and Pharmaceutical Chemistry, 12th Edition, Lippincott Williams & Wilkins, Wolters Kluwer (India) Pvt.Ltd, New Delhi.
3. Comprehensive Medicinal Chemistry – Corwin and Hansch.
4. Computational and structural approaches to drug design edited by Robert M Stroud and Janet. F Moore 81.
5. Introduction to Quantitative Drug Design by Y.C. Martin.
6. Principles of Medicinal Chemistry by William Foye, 7th Edition, Lippincott Williams & Wilkins, Wolters Kluwer (India) Pvt.Ltd, New Delhi.
7. Drug Design Volumes by Arienes, Academic Press, Elsevier Publishers, Noida, Uttar Pradesh.
8. Principles of Drug Design by Smith.

9. The Organic Chemistry of the Drug Design and Drug action by Richard B. Silverman, II Edition, Elsevier Publishers, New Delhi.
10. An Introduction to Medicinal Chemistry, Graham L.Patrick, III Edition, Oxford University Press, USA.
11. Biopharmaceutics and pharmacokinetics, D.M. Brahmkar, Sunil B. Jaiswal II Edition, 2014, Vallabh Prakashan, New Delhi.
12. Peptidomimetics in Organic and Medicinal Chemistry by Antonio Guarna and Andrea Trabocchi, First edition, Wiley publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM003: CHEMISTRY OF NATURAL PRODUCTS (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Study of Natural products as leads for new pharmaceuticals for the following class of drugs	12
2	Alkaloids, Flavonoids and Steroids	12
3	Terpenoids and Vitamins	12
4	Recombinant DNA technology and examples of crude drug use in Diabetic therapy	12
5	Structural elucidation of natural compounds	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Study of Natural products as leads for new pharmaceuticals for the following class of drugs	12	20%
	a) Drugs Affecting the Central Nervous System: Morphine Alkaloids b) Anticancer Drugs: Paclitaxel and Docetaxel, Etoposide, and Teniposide c) Cardiovascular Drugs: Lovastatin, Teprotide and Dicoumarol d) Neuromuscular Blocking Drugs: Curare alkaloids e) Anti-malarial drugs and Analogues f) Chemistry of macrolide antibiotics (Erythromycin, Azithromycin, Roxithromycin, and Clarithromycin) and β - Lactam antibiotics (Cephalosporins and Carbapenem)		

2.	Alkaloids, Flavonoids and Steroids	12	20%
	General introduction, classification, isolation, purification, molecular modification and biological activity of alkaloids, general methods of structural determination of alkaloids, structural elucidation and stereochemistry of ephedrine, morphine, ergot, emetine and reserpine. Introduction, isolation and purification of flavonoids, General methods of structural determination of flavonoids; Structural elucidation of quercetin. General introduction, chemistry of sterols, sapogenin and cardiac glycosides. Stereochemistry and nomenclature of steroids, chemistry of contraceptive agents male & female sex hormones (Testosterone, Estradiol, Progesterone), adrenocorticoids (Cortisone), contraceptive agents and steroids (Vit – D).		
3.	Terpenoids and Vitamins	12	20%
	Classification, isolation, isoprene rule and general methods of structural elucidation of Terpenoids; Structural elucidation of drugs belonging to mono- (citral, menthol, camphor), di-(retinol, Phytol, taxol) and tri- terpenoids (Squalene, Ginsenoside) carotinoids (β carotene). Chemistry and Physiological significance of Vitamin A, B₁, B₂, B₁₂, C, E, Folic acid and Niacin.		
4.	Recombinant DNA technology and examples of crude drug use in Diabetic therapy	12	20%
	4.1 rDNA technology, hybridoma technology, New pharmaceuticals derived from biotechnology; Oligonucleotide therapy. Gene therapy: Introduction, Clinical application and recent advances in gene therapy, principles of RNA & DNA estimation 4.2 Active constituent of certain crude drugs used in Indigenous system Diabetic therapy – <i>Gymnema sylvestre</i>, <i>Salacia reticulata</i>,		

	<i>Pterocarpus marsupium</i> , <i>Swertia chirata</i> , <i>Trigonella foenum gracum</i> ; Liver dysfunction – <i>Phyllanthus niruri</i> ; Antitumor – <i>Curcuma longa</i> Linn.		
5.	Structural elucidation of natural compounds	12	20%
	Structural characterization of natural compounds using IR, ¹ HNMR, ¹³ CNMR and MS Spectroscopy of specific drugs e.g., Penicillin, Morphine, Camphor, Vit-D, Quercetin and Digitalis glycosides.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize different types of natural compounds and their chemistry and medicinal importance.
CO2	describe the importance of natural compounds as lead molecules for new drug discovery.
CO3	comprehend the concept of rDNA technology tool for new drug discovery.
CO4	narrate general methods of structural elucidation of compounds of natural origin.
CO5	understand isolation, purification and characterization of simple chemical constituents from natural source.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ Text book:

1. Modern Methods of Plant Analysis, Peech and M.V.Tracey, Springer – Verlag, Berlin, Heidelberg.
2. Phytochemistry Vol. I and II by Miller, Jan Nostrant Rein Hld.
3. Recent advances in Phytochemistry Vol. I to IV – Scikel Runeckles, Springer Science & Business Media.
4. Chemistry of natural products Vol I onwards IWPAC.
5. Natural Product Chemistry Nakanishi Gggolo, University Science Books, California.
6. Natural Product Chemistry “A laboratory guide” – Rapheal Khan.
7. The Alkaloid Chemistry and Physiology by RHF Manske, Academic Press.
8. Introduction to molecular Phytochemistry – CHJ Wells, Chapmanstall.
9. Organic Chemistry of Natural Products Vol I and II by Gurdeep and Chatwall, Himalaya Publishing House.
10. Organic Chemistry of Natural Products Vol I and II by O.P. Agarwal, Krishan Prakashan.
11. Organic Chemistry Vol I and II by I.L. Finar, Pearson education.
12. Elements of Biotechnology by P.K. Gupta, Rastogi Publishers.
13. Pharmaceutical Biotechnology by S.P.Vyas and V.K.Dixit, CBS Publishers.
14. Biotechnology by Purohit and Mathur, Agro-Bios, 13th edition.
15. Phytochemical methods of Harborne, Springer, Netherlands.
16. Burger’s Medicinal Chemistry.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM004: PHARMACEUTICAL CHEMISTRY PRACTICAL - I
(Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Analysis of Pharmacopoeial compounds and their formulations by UV Vis spectrophotometer, RNA and DNA estimation.
2	Simultaneous estimation of multi component containing formulations by UV spectrophotometry.
3	Experiments based on Column chromatography.
4	Experiments based on HPLC.
5	Experiments based on Gas Chromatography.
6	Estimation of riboflavin/quinine sulphate by fluorimetry.
7	Estimation of sodium/potassium by flame photometry.
8	Purification of organic solvents, column chromatography.
9	To perform the following reactions of synthetic importance Claisen-schmidt reaction Benzyllic acid rearrangement. Beckmann rearrangement. Hoffmann rearrangement Mannich reaction
10	Synthesis of medically important compounds involving more than one step along with purification and Characterization using TLC, melting point and IR spectroscopy.
11	Estimation of elements and functional groups in organic natural compounds
12	Isolation, characterization like melting point, mixed melting point, molecular weight determination, functional group analysis, co-chromatographic technique for identification of isolated compounds and interpretation of UV and IR data.
13	Some typical degradation reactions to be carried on selected plant constituents.

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	perform and optimize medicinally important name reaction.
CO2	purify and characterize the medicinally important synthesized compounds
CO3	perform the analysis of drug/formulation using analytical instruments like HPTLC, HPLC, UV/vis spectrometer etc.
CO4	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	-	3	-	-	-	-	-	3	-
CO2	3	3	-	-	-	-	-	-	-	3	3
CO3	3	3	-	3	-	-	-	-	-	3	3
CO4	-	3	3	-	3	-	-	-	-	-	-

Recommended Study Material:**❖ Text book:**

1. “Advanced Organic chemistry, Reaction, Mechanisms and Structure”, J March, John Wiley and Sons, New York.
2. “Mechanism and Structure in Organic Chemistry”, ES Gould, Hold Rinchart and Winston, New York.
3. “Organic Chemistry” Clayden, Greeves, Warren and Wothers., Oxford University Press 2001.
4. “Organic Chemistry” Vol I and II. I.L. Finar. ELBS, Pearson Education Lts, Dorling Kindersley 9 India) Pvt. Ltd.
5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
6. Reactive Intermediates in Organic Chemistry, Tandon and Gowell, Oxford & IBH Publishers.
7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.

8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wiley India
10. Principles of Organic Synthesis, R.C. Norman and J.M. Coxon, Nelson Thornes.
11. Organic Synthesis - Special Techniques. VK Ahluwalia and R Agarwal, Narosa Publishers.
12. Organic Reaction Mechanisms IVth Edn, VK Ahluwalia and RK Parashar, Narosa Publishers

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS105.02 B: ACADEMIC SPEAKING AND PRESENTATION SKILLS (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/ Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS105.02 B	Academic Speaking and Presentation Skills	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Foundations of Advance Communication - Meaning and Definition of Advance Communication - Advance Communication in Digital, Social, Mobile World - Strategies for Advance Communication - Meaning and Concept of Academic Language - High Frequency Academic Vocabulary	04
2	Art of Conversation - Describing people, places and things - Expressing opinions - Making suggesting - Persuading someone	06

	• <i>Interpreting and Summarizing</i>	
3	<i>Science of Power Speaking</i> • <i>Phonemes</i> • <i>Word Stress</i> • <i>Pronunciation</i> • <i>Intonation</i> • <i>Pause</i> • <i>Register</i> • <i>Fluency</i> • <i>Prosody</i> • <i>Lexical Range</i>	06
4	<i>Academic Speaking Application – Part I</i> • <i>Art of Oratory</i> • <i>Formal Presentation</i> • <i>Speech Analysis – Decoding Best Speeches</i>	08
5	<i>Academic Speaking Application – Part II</i> • <i>Job Interview</i> • <i>Group Discussion</i> • <i>Meeting</i>	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation.

The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

After completion of the course the student would:

CO1	understand and demonstrate advance communication skills and academic speaking.
CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	3	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	2	-	-	3	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester II

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM006: ADVANCE SPECTRAL ANALYSIS (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	UV and IR spectroscopy	12
2	NMR spectroscopy	12
3	Mass Spectroscopy	12
4	Chromatography	12
5	Miscellaneous analysis	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	UV and IR spectroscopy	12	20%
	Wood ward – Fieser rule for 1,3- butadienes, cyclic dienes and α , β -carbonyl compounds and interpretation compounds of enones. ATR-IR, IR Interpretation of organic compounds.		
2.	NMR spectroscopy	12	20%
	1-D and 2-D NMR, NOESY and COSY, HECTOR, INADEQUATE techniques, Interpretation of organic compounds.		
3.	Mass Spectroscopy	12	20%
	Mass fragmentation and its rules, Fragmentation of important functional groups like alcohols, amines, carbonyl groups and alkanes, Meta stable ions, Mc Lafferty rearrangement, Ring rule, Isotopic peaks, Interpretation of organic compounds.		
4.	Chromatography	12	20%
	Principle, Instrumentation and Applications of the following : a)		

	GC-MS b) GC-AAS c) LC-MS d) LC-FTIR e) LC-NMR f) CE-MS g) High Performance Thin Layer chromatography h) Super critical fluid chromatography i) Ion Chromatography j) I-EC (Ion-Exclusion Chromatography) k) Flash chromatography		
5.	Miscellaneous analysis	12	20%
	a) Thermal methods of analysis Introduction, principle, instrumentation and application of DSC, DTA and TGA. b). Raman Spectroscopy Introduction, Principle, Instrumentation and Applications. c). Radio immuno assay Biological standardization, bioassay, ELISA, Radioimmuno assay of digitalis and insulin.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	interpret NMR, Mass and IR spectra of various organic compounds
CO2	understand theoretical and practical skills of the hyphenated instruments
CO3	identify organic compounds by spectral analysis and assay

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ Text book:

1. Spectrometric Identification of Organic compounds - Robert M Silverstein, Sixth edition, John Wiley & Sons, 2004.
2. Principles of Instrumental Analysis - Douglas A Skoog, F. James Holler, Timothy A. Nieman, 5th edition, Eastern press, Bangalore, 1998.

3. Instrumental methods of analysis – Willards, 7th edition, CBS publishers.
4. Organic Spectroscopy - William Kemp, 3rd edition, ELBS, 1991.
5. Quantitative analysis of Pharmaceutical formulations by HPTLC - P D Sethi, CBS Publishers, New Delhi.
6. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi, 3rd Edition, CBS Publishers, New Delhi, 1997.
7. Pharmaceutical Analysis- Modern methods – Part B - J W Munson, Volume 11, Marcel Dekker Series.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM007: ADVANCED ORGANIC CHEMISTRY – II (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Green Chemistry	12
2	Chemistry of Peptides	12
3	Photochemical and Pericyclic Reactions	12
4	Catalysis	12
5	Stereochemistry & Asymmetric Synthesis	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Green Chemistry	12	20%
	a. Introduction, principles of green chemistry b. Microwave assisted reactions: Merit and demerits of its use, increased reaction rates, mechanism, superheating effects of microwave, effects of solvents in microwave assisted synthesis, microwave technology in process optimization, its applications in various organic reactions and heterocycles synthesis c. Ultrasound assisted reactions: Types of sonochemical reactions, homogenous, heterogeneous liquid-liquid and liquid-solid reactions, synthetic applications d. Continuous flow reactors: Working principle, advantages and synthetic applications.		
2.	Chemistry of Peptides	12	20%
	a. Coupling reactions in peptide synthesis		

	b. Principles of solid phase peptide synthesis, t-BOC and Fmoc protocols, various solid supports and linkers: Activation procedures, peptide bond formation, deprotection and cleavage from resin, low and high HF cleavage protocols, formation of free peptides and peptide amides, purification and case studies, site-specific chemical modifications of peptides c. Segment and sequential strategies for solution phase peptide synthesis with any two case studies d. Side reactions in peptide synthesis: Deletion peptides, side reactions initiated by proton abstraction, protonation, over activation and side reactions of individual amino acids.		
3.	Photochemical and Pericyclic Reactions	12	20%
	a. Basic principles of photochemical reactions. Photo-oxidation, photo-addition and photo-fragmentation. b. Mechanism, Types of pericyclic reactions such as cyclo addition, electrocyclic reaction and sigmatropic rearrangement reactions with examples.		
4.	Catalysis	12	20%
	a. Types of catalysis, heterogeneous and homogenous catalysis, advantages and disadvantages. b. Heterogeneous catalysis – preparation, characterization, kinetics, supported catalysts, catalyst deactivation and regeneration, some examples of heterogeneous catalysis used in synthesis of drugs. c. Homogenous catalysis, hydrogenation, hydroformylation, hydrocyanation, Wilkinson catalysts, chiral ligands and chiral induction, Ziegler-Natta catalysts, some examples of homogenous catalysis used in synthesis of drugs. d. Transition-metal and Organo-catalysis in organic synthesis: Metal-catalyzed reactions e. Biocatalysis: Use of enzymes in organic synthesis,		

	immobilized enzymes/cells in organic reaction. f. Phase transfer catalysis - theory and applications.		
5.	Stereochemistry & Asymmetric Synthesis	12	20%
	Basic concepts in stereochemistry – optical activity, specific rotation, racemates and resolution of racemates, the Cahn, Ingold, Prelog (CIP) sequence rule, meso compounds, pseudo asymmetric centres, axes of symmetry, Fischers D and L notation, cis-trans isomerism, E and Z notation. Methods of asymmetric synthesis using chiral pool, chiral auxiliaries and catalytic asymmetric synthesis, enantiopure separation and Stereo selective synthesis with examples.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe the principles and applications of Green chemistry
CO2	understand the concept of peptide chemistry
CO3	summarize the various catalysts used in organic reactions
CO4	comprehend concept of stereochemistry and asymmetric synthesis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ Text book:

1. “Advanced Organic chemistry, Reaction, mechanisms and structure”, J March, John Wiley and sons, New York.

2. "Mechanism and structure in organic chemistry", ES Gould, Hold Rinchart and Winston, New York.
3. "Organic Chemistry" Clayden, Greeves, Warren and Wothers. Oxford University Press 2001.
4. "Organic Chemistry" Vol I and II. I.L. Finar. ELBS, Sixth ed., 1995.
5. Carey, Organic chemistry, 5th edition (Viva Books Pvt. Ltd.)
6. Organic synthesis-the disconnection approach, S. Warren, Wiley India
7. Principles of organic synthesis, R O C Norman and J M Coxan.
8. Organic synthesis- Special techniques VK Ahluwalia and R Aggarwal, Narosa Publishers.
9. Organic reaction mechanisms IVth edition, VK Ahluwalia and RK Parashar, Narosa Publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM008: COMPUTER AIDED DRUG DESIGN (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Quantitative Structure Activity Relationships-Basic	12
2.	Quantitative Structure Activity Relationships-Application	12
3.	Molecular Modeling and Docking	12
4.	Molecular Properties and Drug Design	12
5.	Pharmacophore Mapping and Virtual Screening	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Quantitative Structure Activity Relationships-Basic	12	20%
	History and development of QSAR: Physicochemical parameters and methods to calculate physicochemical parameters: Hammett equation and electronic parameters (sigma), lipophilicity effects and parameters (log P, pi-substituent constant), steric effects (Taft steric and MR parameters) Experimental and theoretical approaches for the determination of these physicochemical parameters.		
2.	Quantitative Structure Activity Relationships: Applications	12	20%
	Hansch analysis, Free Wilson analysis and relationship between them, Advantages and disadvantages; Deriving 2D-QSAR equations. 3D-QSAR approaches and contour map analysis. Statistical methods used in QSAR analysis and importance of statistical parameters.		

3.	Molecular Modeling and Docking	12	20%
	a) Molecular and Quantum Mechanics in drug design. b) Energy Minimization Methods: comparison between global minimum conformation and bioactive conformation c) Molecular docking and drug receptor interactions: Rigid docking, flexible docking and extra-precision docking. Agents acting on enzymes such as DHFR, HMG-CoA reductase and HIV protease, choline esterase (AChE & BChE)		
4.	Molecular Properties and Drug Design	12	20%
	a) Prediction and analysis of ADMET properties of new molecules and its importance in drug design. b) De novo drug design: Receptor/enzyme-interaction and its analysis, Receptor/enzyme cavity size prediction, predicting the functional components of cavities, Fragment based drug design. c) Homology modeling and generation of 3D-structure of protein		
5.	Pharmacophore Mapping and Virtual Screening	12	20%
	a. Concept of pharmacophore, pharmacophore mapping, identification of Pharmacophore features and Pharmacophore modeling; Conformational search used in pharmacophore mapping b. <i>In Silico</i> Drug Design and Virtual Screening Techniques Similarity based methods and Pharmacophore based screening, structure based In-silico virtual screening protocols.		

Course Outcome (COs):

At the end of the course, the students would be able to;

Role of CADD in drug discovery • Different CADD techniques and their applications • Various strategies to design and develop new drug like molecules. • Working with molecular modeling softwares to design new drug molecules • The in silico virtual screening protocols

CO1	understand role of CADD in drug discovery.
CO2	summarize different CADD techniques and their applications
CO3	comprehend various strategies to design and develop new drug like molecules
CO4	describe in silico virtual screening protocols and molecular modeling softwares.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	3	-	-	-	-	-	-	3

Recommended Study Material:

❖ Textbook:

1. Computational and structural approaches to drug discovery, Robert M Stroud and Janet. F Moore, RCS Publishers.
2. Introduction to Quantitative Drug Design by Y.C. Martin, CRC Press, Taylor & Francis group.
3. Drug Design by Ariens Volume 1 to 10, Academic Press, 1975, Elsevier Publishers.
4. Principles of Drug Design by Smith and Williams, CRC Press, Taylor & Francis.
5. The Organic Chemistry of the Drug Design and Drug action by Richard B. Silverman, Elsevier Publishers.
6. Medicinal Chemistry by Burger, Wiley Publishing Co.
7. An Introduction to Medicinal Chemistry –Graham L. Patrick, Oxford University Press.
8. Wilson and Gisvold's Text book of Organic Medicinal and Pharmaceutical Chemistry, Ippincott Williams & Wilkins.
9. Comprehensive Medicinal Chemistry – Corwin and Hansch, Pergamon Publishers.
10. Computational and structural approaches to drug design edited by Robert M Stroud and Janet. F Moore.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM009: PHARMACEUTICAL PROCESS CHEMISTRY (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Process Chemistry	12
2	Unit operations	12
3	Unit Processes – I	12
4	Unit Processes – II	12
5	Industrial Safety	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	Process Chemistry	12	20%
	Introduction, Synthetic strategy Stages of scale up process: Bench, pilot and large scale process. In-process control and validation of large scale process. Case studies of some scale up process of APIs. Impurities in API, types and their sources including genotoxic impurities		
2.	Unit operations	12	20%
	a) Extraction: Liquid equilibria, extraction with reflux, extraction with agitation, counter current extraction. b) Filtration: Theory of filtration, pressure and vacuum filtration, centrifugal filtration. c) Distillation: azeotropic and steam distillation d) Evaporation: Types of evaporators, factors affecting evaporation.		

	e) Crystallization: Crystallization from aqueous, non-aqueous solutions factors affecting crystallization, nucleation. Principle and general methods of Preparation of polymorphs, hydrates, solvates and amorphous APIs.		
3.	Unit Processes – I	12	20%
	a) Nitration: Nitrating agents, Aromatic nitration, kinetics and mechanism of aromatic nitration, process equipment for technical nitration, mixed acid for nitration. b) Halogenation: Kinetics of halogenations, types of halogenations, catalytic halogenations. Case study on industrial halogenation process. c) Oxidation: Introduction, types of oxidative reactions, Liquid phase oxidation with oxidizing agents. Non-metallic Oxidizing agents such as H₂O₂, sodium hypochlorite, Oxygen gas, ozonolysis.		
4.	Unit Processes – II	12	20%
	a) Reduction: Catalytic hydrogenation, Heterogeneous and homogeneous catalyst; Hydrogen transfer reactions, Metal hydrides. Case study on industrial reduction process. b) Fermentation: Aerobic and anaerobic fermentation. Production of i. Antibiotics; Penicillin and Streptomycin, ii. Vitamins: B2 and B12 iii. Statins: Lovastatin, Simvastatin c) Reaction progress kinetic analysis i. Streamlining reaction steps, route selection, ii. Characteristics of expedient routes, characteristics of cost-effective routes, reagent selection, families of reagents useful for scale-up.		
5.	Industrial Safety	12	20%
	a) MSDS (Material Safety Data Sheet), hazard labels of chemicals and Personal Protection Equipment (PPE) b) Fire hazards, types of fire & fire extinguishers c) Occupational Health & Safety Assessment Series 1800		

	(OHSAS-1800) and ISO-14001(Environmental Management System), Effluents and its management		
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Course Outcome (COs):

At the end of the course, the students would be able to;

The strategies of scale up process of APIs and intermediates • The various unit operations and various reactions in process chemistry

CO1	describe the strategies of scale up process of APIs and intermediates.
CO2	summarize various unit operations and various reactions in process chemistry.
CO3	comprehend hazard and safety management of chemical and equipment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	3	3

Recommended Study Material:

❖ Textbook:

1. Process Chemistry in the Pharmaceutical Industry: Challenges in an Ever Changing Climate-An Overview; K. Gadamasetti, CRC Press.
2. Pharmaceutical Manufacturing Encyclopedia, 3rd edition, Volume 2.
3. Medicinal Chemistry by Burger, 6th edition, Volume 1-8.
4. W.L. McCabe, J.C Smith, Peter Harriott. Unit operations of chemical engineering, 7th edition, McGraw Hill
5. Polymorphism in Pharmaceutical Solids .Dekker Series Volume 95 Ed: H G Brittain (1999)
6. Regina M. Murphy: Introduction to Chemical Processes: Principles, Analysis, Synthesis

7. Peter J. Harrington: Pharmaceutical Process Chemistry for Synthesis: Rethinking the Routes to Scale-Up
8. P.H. Groggins: Unit processes in organic synthesis (MGH)
9. F.A. Henglein: Chemical Technology (Pergamon)
10. M. Gopal: Dryden's Outlines of Chemical Technology, WEP East-West Press
11. Clausen, Mattson: Principle of Industrial Chemistry, Wiley Publishing Co.,
12. Lowenheim & M.K. Moran: Industrial Chemicals
13. S.D. Shukla & G.N. Pandey: A text book of Chemical Technology Vol. II, Vikas Publishing House
14. J.K. Stille: Industrial Organic Chemistry (PH)
15. Shreve: Chemical Process, Mc Grawhill.
16. B.K. Sharma: Industrial Chemistry, Goel Publishing House
17. ICH Guidelines
18. United States Food and Drug Administration official website www.fda.gov

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM010: PHARMACEUTICAL CHEMISTRY PRACTICAL-II
(Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Synthesis of organic compounds by adapting different approaches involving (3 experiments) a) Oxidation b) Reduction/hydrogenation c) Nitration
2	Comparative study of synthesis of APIs/intermediates by different synthetic routes (2 experiments)
3	Assignments on regulatory requirements in API (2 experiments)
4	Comparison of absorption spectra by UV and Wood ward – Fieser rule
5	Interpretation of organic compounds by FT-IR.
6	Interpretation of organic compounds by MASS
7	Interpretation of organic compounds by NMR
8	Determination of purity by DSC in pharmaceuticals
9	Identification of organic compounds using FT-IR, NMR, CNMR and Mass spectra.
10	To carry out the preparation of following organic compounds a. Preparation of 4-chlorobenzhydrylpiperazine. (an intermediate for cetirizine HCl). b. Preparation of 4-iodotoluene from p-toluidine. c. NaBH ₄ reduction of vanillin to vanillyl alcohol d. Preparation of umbelliferone by Pechhman reaction e. Preparation of triphenyl imidazole
11	To perform the Microwave irradiated reactions of synthetic importance (Any two) such as phenytoin and Biginellii compound.
12	Determination of log P, MR, hydrogen bond donors and acceptors of selected drugs using software
13	Calculation of ADMET properties of drug molecules and its analysis using software.

14	Pharmacophore modelling based experiment
15	2D-QSAR and 3D QSAR based experiments
16	Docking study and virtual screening based experiment

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	elucidate organic compound by spectroscopic techniques.
CO2	synthesize various APIs and intermediates by conventional and green chemistry approach
CO3	perform and interpret in silico ADMET properties of organic compounds
CO4	perform virtual screening and pharmacophore based experiment
CO5	analyze the problem, communicate suggested solution and interpret the results

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	3	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	3	3	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	3	-	-	3

Recommended Study Material:

❖ Textbook:

1. “Advanced Organic chemistry, Reaction, Mechanisms and Structure”, J March, John Wiley and Sons, New York.
2. “Mechanism and Structure in Organic Chemistry”, ES Gould, Hold Rinchart and Winston, New York.
3. “Organic Chemistry” Clayden, Greeves, Warren and Wothers., Oxford University Press 2001.
4. “Organic Chemistry” Vol I and II. I.L. Finar. ELBS, Pearson Education Lts, Dorling Kindersley 9 India) Pvt. Ltd.
5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
6. Reactive Intermediates in Organic Chemistry, Tandon and Gowen, Oxford & IBH Publishers.
7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.

8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wiley India
10. Principles of Organic Synthesis, R.C. Norman and J.M. Coxon, Nelson Thornes.

**CHAROTAR UNIVERSITY OF SCIENCE &
TECHNOLOGY**
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS106.02 B: ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
II	HS106.02 B	Academic Writing	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	Academic Writing and Research Process • <i>Introduction to Academic Writing</i> • <i>Academic Writing as a Part of Research</i> • <i>Types of Academic Writing</i> • <i>Features of Academic Writing</i> • <i>Importance of Good Academic Writing in various Academic Works</i>	05
2	Anatomy of Academic Writing • <i>Academic Vocabulary</i> • <i>Simple and Complex Sentences</i> • <i>Organizing Paragraphs</i> • <i>The Writing Process</i> • <i>Adopting Academic Writing Style</i>	05
3	Key Academic Skills • <i>Note – taking</i>	05

	• Note – making • Paraphrasing • Summarizing	
4	Accuracy in Academic Writing • <i>Lexical Range</i> • <i>Academic Language and Structures</i> • <i>Elements of Writing</i> • <i>Proof Reading, Editing, and Rewriting</i>	05
5	Using and Citing Sources of Ideas • <i>Academic Texts and their Types</i> • <i>Intellectual Honesty in Academic Writing</i> • <i>Avoiding Plagiarism – Idea Theft</i> • <i>Degrees of Plagiarism</i> • <i>Types of Borrowing</i> • <i>Anatomy of Citations</i> • <i>Common Citation Styles</i>	05
6	Contemporary Practices in Academic Writing • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	05
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading**Essential Reading for Concepts**

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- *Effective Academic Writing Level - 1,2,3,4 (Second Edition)* By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu.

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester III

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC010 : RESEARCH METHODOLOGY & BIOSTATISTICS

Total hours: 60

UNIT – I General Research Methodology: **12 hr**

Research, objective, requirements, practical difficulties, review of literature, study design, types of studies, strategies to eliminate errors/bias, controls, randomization, crossover design, placebo, blinding techniques.

UNIT – II Biostatistics: **12 hr**

Definition, application, sample size, importance of sample size, factors influencing sample size, dropouts, statistical tests of significance, type of significance tests, parametric tests (students “t” test, ANOVA, Correlation coefficient, regression), non-parametric tests (wilcoxon rank tests, analysis of variance, correlation, chi square test), null hypothesis, P values, degree of freedom, interpretation of P values.

UNIT – III Medical Research: **12 hr**

History, values in medical ethics, autonomy, beneficence, non-maleficence, double effect, conflicts between autonomy and beneficence/non-maleficence, euthanasia, informed consent, confidentiality, criticisms of orthodox medical ethics, importance of communication, control resolution, guidelines, ethics committees, cultural concerns, truth telling, online business practices, conflicts of interest, referral, vendor relationships, treatment of family members, sexual relationships, fatality.

UNIT – IV CPCSEA guidelines for laboratory animal facility: **12 hr**

Goals, veterinary care, quarantine, surveillance, diagnosis, treatment and control of disease, personal hygiene, location of animal facilities to laboratories, anesthesia, euthanasia, physical facilities, environment, animal husbandry, record keeping, SOPs, personnel and training, transport of lab animals.

UNIT – V Declaration of Helsinki: **12 hr**

History, introduction, basic principles for all medical research, and additional principles for medical research combined with medical care.

A. Recommended Study Material:

❖ Text Books:

1. Research Methodology, Methods & Techniques, C.R. Kothari, Viswa Prakashan, 2nd Edition, 2009.
2. Research Methods- A Process of Inquiry, Graziano, A.M., Raulin, M.L, Pearson Publications, 7th Edition, 2009.
3. How to Write a Thesis:, Murray, R. Tata McGraw Hill, 2nd Edition, 2010.
4. Writing For Academic Journals, Murray, R., McGraw Hill International, 2009.
5. Writing for Publication, Henson, K.T., Allyn & Bacon, 2005.

❖ Reference Books:

1. What is this thing called Science, Chalmers, A.F., Queensland University Press, 1999.
2. Methods & Techniques of Social Research, Bhandarkar & Wilkinson, Himalaya publications, 2009.
3. Doing your Research project, Bell J., Open University Press, Berkshire, 4th Edition, 2005
4. A Handbook of Academic Writing, Murray, R. and Moore, S., Tata McGraw Hill International, 2006.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM013: RESEARCH WORK-I

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

**FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC011: JOURNAL CLUB -I**

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester IV

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM015 : RESEARCH WORK-II

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

**FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC012: JOURNAL CLUB -II**

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3